

# Experimental Toolkit Procedure - Photosynthesis and Plant Efficiency for Bioregenerative Life Support Systems

## GENERAL INFORMATION

|                |                          |
|----------------|--------------------------|
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## SHORT EXPERIMENT DESCRIPTION

The objective of the experimental kit “Photosynthesis and Plant Efficiency for Bioregenerative Life Support Systems” is to teach the fundamentals of photosynthesis in an exciting, hands-on way, and to connect this core biology topic to real-world applications in Bio-regenerative Life Support Systems for human spaceflight. Students work in four groups, each assigned a different lighting condition (natural light, complete darkness, green-filtered light, or red-filtered light), and investigate how these conditions affect the CO<sub>2</sub> concentration inside a sealed jar containing a basil plant. For documentation, each student fills out a worksheet with their hypothesis, procedure, observations, and evaluation, supported throughout by a set of student flashcards.

It is foreseen that the execution of this experiment takes around 45 minutes.

## HARDWARE CHECKLIST

### *From the school/students:*

|  |                                                             |
|--|-------------------------------------------------------------|
|  | 2x Healthy basil plants                                     |
|  | Personal supplies for each student: pencil, ruler, notebook |
|  | Timer (e.g. smartphone stopwatch)                           |

### *Included in the kits:*

|  |                                                                  |
|--|------------------------------------------------------------------|
|  | Power supply or power bank (USB Type A)                          |
|  | Tape                                                             |
|  | 4 x Airtight glass containers (jar) with a small hole in the lid |
|  | Coloured filters in red and green                                |
|  | Aluminum foil                                                    |
|  | 4 x CO <sub>2</sub> sensor (incl. cable)                         |
|  | Student Flashcards (1 set per group)                             |
|  | Worksheet / Lab Report (1 per student)                           |

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|  | Teacher's Script |
|--|------------------|

## PROCEDURE

The students shall use the provided flashcards to gain background knowledge and the provided worksheet to summarize the experiment's key findings. In addition, the teacher can use the teacher's script to gain a deeper understanding of the background knowledge.

### Set Up the Experiment

| Step | Action                                                                                                                                                               | Notes                                                                                                                 | Duration | Check |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------|-------|
| 1    | Divide the class into four groups.                                                                                                                                   | Assign each group a lighting condition: Complete Darkness, Green-Filtered Light, or Red-Filtered Light, Natural Light | 3 min.   |       |
| 2    | Distribute one jar and one CO <sub>2</sub> sensor to each of the four groups. Make sure that each group has access to the online available worksheet and flashcards. |                                                                                                                       | 2 min.   |       |
| 3    | Distribute to three of the four groups the group-specific material                                                                                                   | The fourth group does not need any additional material                                                                |          |       |
| 3.1  | Distribute to the first Group the aluminum foil                                                                                                                      |                                                                                                                       | 1 min.   |       |
| 3.2  | Distribute to the second group the green filter                                                                                                                      |                                                                                                                       | 1 min.   |       |
| 3.3  | Distribute to the third group the red filter                                                                                                                         |                                                                                                                       | 1 min.   |       |
| 4    | Each group prepares the jar according to the assigned lighting condition.                                                                                            | The preparations shall be done in parallel                                                                            | 5 min.   |       |

|     |                                                                                                                                   |                                                                                      |  |  |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--|--|
| 4.1 | Group 1 wraps the jar in aluminum foil, ensuring that no illumination reaches the inside of the jar                               |    |  |  |
| 4.2 | Group 2 wraps the jar in the green filter, ensuring that no illumination reaches the inside of the jar without passing the filter |   |  |  |
| 4.3 | Group 3 wraps the jar in the red filter, ensuring that no illumination reaches the inside of the jar without passing the filter   |  |  |  |

|     |                                                                                                                                                      |                                                                                                                     |               |  |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------|--|
| 4.4 | Group 4 does not adapt the jar in any way                                                                                                            |                                   |               |  |
| 5   | Each group: Thread the CO <sub>2</sub> sensor cable through the small aperture in the lid and seal the opening with tape.                            |                                   | 1 min.        |  |
| 6   | Each group: Connect the sensor to the power supply or power bank.                                                                                    |                                                                                                                     | A few seconds |  |
| 7   | Each group: Record the baseline CO <sub>2</sub> concentration.                                                                                       | Place the sensor inside the empty, sealed jar and note the CO <sub>2</sub> level once it stays within $\pm 15$ ppm. | 4 min.        |  |
| 8   | Each group: While waiting for the CO <sub>2</sub> baseline to stabilize, cut the stems and leaves of half of one basil plant pot (no roots or soil). |                                                                                                                     |               |  |
| 9   | Each group: Open the jar and place the prepared basil inside.                                                                                        |                                 | 1 min.        |  |
| 10  | Each group: Tightly close the lid and start the timer.                                                                                               | The timer should count up until 12 min.                                                                             | A few seconds |  |
| 11  | Each group: Until the                                                                                                                                | Keep the jar sealed and                                                                                             | 12 min.       |  |

|  |                                                                                          |                                                                                               |  |  |
|--|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--|--|
|  | timer reaches 12 min.<br>record the CO <sub>2</sub><br>concentration every 2<br>minutes. | undisturbed, and keep all<br>setups in the same lighting<br>environment for<br>comparability. |  |  |
|--|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--|--|

### Experiment Post-Processing

|   |                                      |                                                                                                                                                                         |         |  |
|---|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--|
| 1 | Complete the worksheet.              | Every student fills in the<br>findings for the lighting<br>conditions they did not<br>personally test, so each<br>worksheet documents all<br>four conditions.           | 10 min. |  |
| 2 | Conclude with a class<br>discussion. | Summarize how light<br>intensity and wavelength<br>affected CO <sub>2</sub><br>concentration, and<br>reinforce that<br>photosynthesis is a light-<br>dependent process. | 2 min.  |  |